ECO TECH DOMICILE

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ABSTRACT

As the world progresses and advances in technology development and uses of technology in solving challenges, Egypt must take serious steps to catch thriving countries. One of Egypt's grand challenges is improving the technological environment. Now, our project will solve some challenges with using technology, such as climate change. The purpose of the project is to reduce climate change by building smart homes with natural materials that make the inner temperature of the home 3 degrees Celsius lower than the outer environment. Palm tree branches are used to build the house and cover it with pottery clay. The house is supported by a smart infrastructure. Sensors and actuators are used to detect and take an action when a condition happens. The Mq2 sensor is used as a gas sensor, the IR sensor as a motion sensor, and two LM35 sensors as temperature sensors. The Mq2 is used to sense any gas or smoke, and then the action is to open windows and turn on a fan to extract air while executing a code for the alert system. An IR sensor is used in the security system to detect any motion during the period of arming the security system. All sensors are tested, and actions occur. All these actions are controlled by the code. The LM35 reads the temperature and sends it via Bluetooth module to a selected smartphone. After finishing constructing and coding, this is the time for the test plan. The test plan goes well, and the design requirements are achieved where the MQ-2 ventilation system deals with any smoke or gas leakage. Security systems maintain a high level of security. The difference between the outside and inside was an average of four degrees. Finally, our project meets the design requirements and solves the specified problems.

INTRODUCTION

Egypt faces a lot of grand challenges (Recycling garbage and waste for economic and environmental purposes, Improving the scientific and technological environment for all, Reducing and adapting to the effects of climatic change and Increasing the industrial and agricultural base of Egypt. First, to deal with these problems, we must know the definition of each problem. First Recycling garbage and waste, waste management is becoming a problem for major cities worldwide. This is especially serious in developing countries because of the rapid increase in SW generation caused by rapid population growth and urbanization. The second scientific and technological environment, the collective panorama of scientific and technological breakthroughs, discoveries, approaches, tools, and infrastructure is referred to as the scientific and technological environment. Third Reduce and adapt to the effect of climatic change, Climate change is a pressing issue of the 21st century, resulting from human activities like fossil fuel burning and deforestation, leading to global warming. Finally Increase the industrial and agricultural base, Industry and agriculture are the most important Economic activities in Egypt and they also have a great role in the national income. All these problems can be solved by our project (a sustainable smart home). There are many prior solutions for this topic. For example, in Burj Kalifa and Nile City towers, smart technologies are commonly integrated into modern mixed-use developments: 1- Building Management System (BMS). 2-Smart Lighting Controls. 3-Energy Management Systems. The building materials for these projects aren't sustainable, on the other hand, the smart system saves a lot of energy. From these prior solutions, we can conclude the solution that is supposed to solve the previous grand challenges. The solution reached is building a smart home with palm tree branches as a natural material covering it with a layer of pottery clay and using solar cells as a renewable source. Using a Bluetooth module connected to the Arduino circuit to send notifications to a certain cell phone. Before we began, we had to Choose the design requirements. So, we chose the following and tried to apply them to the project. The design requirements are decreasing the Temperature by 3 Celsius or more from outside the home, high level of security by implementing a specific security system and maintaining a very low level of smoke by using a smoke leakage system. But does our solution address these design requirements? Of course, yes where We achieve the first design requirement by making a clay layer outside the home to work as an isolation layer to decrease the temperature. we achieve the second design requirement by making a security system. The security system depends on arming and disarming the system and the detection of the IR sensor. Finally, the third design requirement. we used a smoke sensor (MQ-2) to detect the smoke. According to the detection, there are some instructions that are going to be done related to the Ventilation system.

MATERIALS

TOTAL COAST : 2200

METHODS

Building and constructing the home itself:

We make some holes in the branch to collect and fix the rest.

We make each side separately then collect the ropes.

We used isolated polycarbonic sheets to construct the windows and doors.

We fixed the hinges on the windows and door.

We cover our home by clay pottery to decrease the temperature inside the home

Connecting components and coding:

We searched for the connections for the components. And started to make these connections in our project.

We started to test every sensor separately to be sure of its functionality.

We started to write the code for each system using IDE, and then we emerge them.

We finally made the test plan for the whole system and building

fig. 1 (the prototype)

TEST PLAN &RESULTS

First, we charge our batteries with solar cells via a charging module. when the batteries become fully charged. We used it to run the Arduino. We uploaded our code to Arduino and tested it by some steps.

1. The user entered a certain code (111) on the keypad, which is located outside the door, to activate security and entered another code to deactivate it (333).

• The security system worked as follows: by putting something near the IR sensor The IR gets a signal, and depending on it, a buzzer will trigger, and a warning notification will be sent to the user.

2. We check the MQ2 sensor. and the sensor gets a fast response and runs the fan to get rid of the smoke or gas and the servo motor to open the window. In addition, there is a warning message to the user, the LED will turn on, and the buzzer will trigger with another tone.

3. We get a read of the internal temperature sensor and the external temperature sensor and represent them on the application.

Cuboid		Volume
length	60 ± 1 cm	72000 ± 1 cm^3
width	30 ± 1 cm	
height	40 ± 1 cm	

Table. 1 (cuboid dimensions)

Triangle prism	Area of base	Height	Volume
	1800 ± 1 cm^2	18.5 ± 1 cm	33300 ± 1 cm^3

Table. 2 (prism dimensions)

Graph. 3 (temperature in Celsius)

	Read 1	Read 2	Read 3	Read 4
Inside	18.3	18.0	19.7	18.1
outside	22.8	22.4	23.1	22.7

Table. 3 (temperature in Celsius)

ANALYSIS

Our project is divided into some partitions: Building the home: We chose some materials that are efficient in isolating the temperature; these materials are palm branches, which will be covered with pottery clay. Natural shade from palm branches can drastically cut down on the amount of direct sunlight that reaches the ground or neighboring structures. Strong winds can be repelled by using palm branches as a natural barrier. They can lower the ambient temperature by obstructing or decelerating air circulation, which can produce pockets of colder air behind them. Because pottery clay has a high thermal mass, heat energy may be efficiently absorbed and stored in it. The clay takes in heat from the surroundings during the day while it's hot outside. This heat is released by the clay at night when the temperature drops, assisting in the maintenance of a more constant interior temperature. Because of its capacity to both absorb and release moisture from the air, clay can help control the humidity levels within buildings. By using pottery clay, you can use fewer synthetic materials, which have a greater impact on the environment. Choosing the suitable components for the circuit: Our home consists of three main systems: the first is a smoke leakage system based on using an MQ-2 smoke sensor to detect any leakage, and accordingly, some instructions are going to be executed (opening the upper window using a servo, turning on a fan to extract air, turning on a buzzer as an alert, blinking a LED, and sending a message to a selected cellphone). The second is a security system using an IR sensor. This idea is based on arming the system by entering a specific code on the keypad, and while the system is activating, the IR sensor detects any motion inside the home, triggers an alert, and sends a message to a cellphone saying that there is a stranger in the home. To disarm the system, another code must be printed on the keypad. The third is using two temperature sensors to measure the temperature outside and inside the home and send the readings to a cellphone. The power supply for the system is: We used solar cells as a source of electricity to charge the battery and use it to power the Arduino. A 30S-20A lithium-ion battery charger is used.

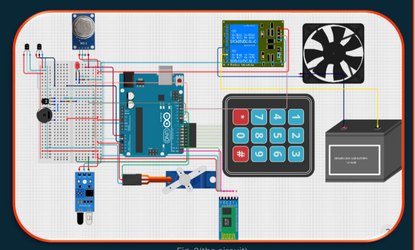

Fig. 2(the circuit)

```
// Read the analog values from the sensors
int sensorValue1 = analogRead(tempInside);
int sensorValue2 = analogRead(tempoutside);

// Convert analog values to temperature in Celsius
float mv1 = (sensorValue1 /1024.0) *500; // LM35 output is 10 mV per degree Celsius
float mv2 = (sensorValue2 /1024.0) *500;
float temperature1 = mv1/10;
  float temperature2 = mv2/10;

// Print the temperatures on the serial monitor
Serial.print("Temperature Sensor outside: ");
Serial.print(temperature1);
Serial.println(" °C");

Serial.print("Temperature Sensor inside: ");
Serial.print(temperature2);
Serial.println(" °C");
// Send temperatures over Bluetooth
bluetooth.print("Temperature outside: ");
bluetooth.print(temperature1);
bluetooth.println(" °C");

bluetooth.print("Temperature Inside: ");
bluetooth.print(temperature2);
bluetooth.println(" °C");
// Wait for a short delay before reading the sensors again
delay(1000);
```

Code. 1(temperature)

This is the code for the temperature sensor to determine the inside and outside temperatures. The variables are defined for the temperature outside and inside the home. sensor Value/1024.0: If the sensor has a 10-bit ADC (Analog-to-Digital Converter) resolution, 1024 is the maximum value, and this normalizes the sensor data to a number between 0 and 1. The normalized value is scaled by multiplying it by 500 to match the output of the LM35 sensor, which produces 10 mV per degree Celsius. Therefore, the normalized value is effectively converted to millivolts by multiplying by 500. float temperature1 = mv1/10: Since the LM35 sensor outputs 10 mV per degree Celsius, dividing mv1 by 10 gives the temperature in degrees Celsius. Bluetooth. Println (temperature): to print the values of the temperature in the cell phone using the HC05 Bluetooth module.

```
int sensorValue = analogRead(mq2Pin); // Read analog value from MQ-2 sensor
if (sensorValue > 350) { // If sensor reading exceeds the threshold
  bluetooth.println("smoke detected"); // Send "Gas detected" over Bluetooth
  Serial.println("smoke detected"); // Print "Gas detected" to serial monitor for debugging
  digitalWrite(fan,HIGH);
  myservo.Write(180);
  digitalWrite(led, HIGH);
  digitalWrite(BUZZER_PIN, HIGH); // Turn on the LED indicator
  delay(5000); // Wait for 10 seconds
  digitalWrite(led, LOW);
  digitalWrite(BUZZER_PIN, LOW);// Turn off the LED indicator
   digitalWrite(fan,LOW);
  myservo.Write(0);
}
else {

  digitalWrite(led, LOW);
  digitalWrite(BUZZER_PIN, LOW);
  digitalWrite(fan,LOW);
  myservo.Write(0);
}
```

Code. 2(smoke leakage system)

The analog value from the MQ-2 sensor attached to pin mq2Pin is read using the formula **int sensor Value = analog Read(mq2Pin)**. The voltage measured by the sensor is returned by the analog Read() function as a value between 0 and 1023. **if (sensor Value > 350)** { Verifies whether the sensor reading is higher than the 350-threshold value. **Bluetooth. println("smoke detected");** Bluetooth transmits the alert "smoke detected." This is probably used to tell another device like a smartphone about the detection.

Activates an LED and buzzer with the code **digital Write(led, HIGH), digital Write(buzzer, HIGH).** And then turning them off if there is no smoke. Also opens the window using servo by **my servo. Write(180)** and turn on a fan by **digital Write(fan, HIGH),** and to turn them off when there is no smoke.

```
void armSystem() {
  if (authenticateUser("111")) {
    systemArmed = true;
    digitalWrite(BUZZER_PIN, HIGH);  // Turn on the buzzer to indicate armed state
    delay(1000); // Delay for one second
    digitalWrite(BUZZER_PIN, LOW); // Turn off the buzzer after one second
    bluetooth.println("System armed");
  } else {
    bluetooth.println("Incorrect arm code");
  }
}

void disarmSystem() {
  if (authenticateUser("333")) {
    systemArmed = false;
    bluetooth.println("System disarmed");
  } else {
    bluetooth.println("Incorrect disarm code");
  }
}
```

Code. 3(security system)

Arming the System: The user enters a specific code using the keypad to arm the system. Once the correct code is entered, the system activates the IR sensor to detect any motion.

Disarming the System: The user enters another specific code to disarm the system. Upon successful entry of the disarm code, the system deactivates the IR sensor.

Intrusion Detection: While the system is armed, if the IR sensor detects motion, it triggers the buzzer to sound an alarm, indicating a potential intrusion.

Functions:

- arm System(): It checks if the user enters the correct code (in this case, "111"). If authenticated, it sets system Armed to true, activates the buzzer briefly, and sends a message via Bluetooth indicating that the system is armed.
- disarm System(): It checks if the user enters the correct disarm code (in this case, "333"). If authenticated, it sets system Armed to false and sends a message via Bluetooth indicating that the system is disarmed.
- authenticate User(String code): This function prompts the user to enter a code via the keypad until the # key is pressed. It then compares the entered code with the provided one and returns true if they match.
- send Alert(): This function sends a message via Bluetooth and activates the buzzer briefly to alert about a detected stranger (motion detected by the IR sensor).

LEARNINGS TRANSFER

Geo.2.09	Study the evolution of Earth, and its layers that enable us to know materials that are stable in high temperature.
ph.2.10	Study AC generators, DC generators, DC motors. To know how can we control the velocity and the power of the actuators we are going to use.
Ph.2.15	We studied about diodes, pn junction, transistors, their working principle and their uses.
Ch.2.09	Study different types of batteries and how it work in charging and discharging and knowing its components.
ENG.2.01	We learned in English about the mind mapping.

CONCLUSION

At the end of the project, it has achieved the predicted results, and all the systems are under control. We used four sensors in the system. The IR sensor detected motion, and it directly gave an action to the buzzer to emit a specific tone. The Mq2 sensor, which was a smoke detector, When Mq2 senses any smoke or gas, it directly administers an action to the buzzer to make a tone different than the IR one. It administers an action to two actuators, which are a fan to get rid of the smell of the smoke and a servo motor to open a window to work as a ventilation supply. The other two sensors are two LM temperature sensors to detect temperatures inside and outside the home. There is another servo motor to open the lock on the home door. We supported the home with a security system where the whole system is armed and disarmed when the user enters the correct password. The whole system is linked to the cellphone via Bluetooth module HC-05. All the previous actions are sent to the cellphone by notification. Finally, this is what our projects have achieved, and we think we succeeded in this project.

RECOMMENDATIONS

While building the prototype, we faced problems with each smart system and the house's structure. I think avoiding these problems can save time and money. Here are some recommendations that can help in making a better project:

1. The first source of information is workshops or experts not only websites or web sources.

2. It's better to buy new electronic materials, use them for the first time, and work efficiently.

3. Building the house with more adhesive material such as concrete that can carry heavy weights and help in inserting the system inside the house

4. Try to reduce the inner temperature as much as possible to provide a safe environment for electronic materials inside the house.

5. Growing plants inside the house is good for providing a low temperature and nice weather inside the house and achieving the main purpose of the project.

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